Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	0x404
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 9 – Industry, Innovation, and Infrastructure
Project / Solution Name	GrantFlow
Name (Leader)	Khoo Zong Hao
Email & Phone (Leader)	zonghao04@gmail.com
University Affiliation(s)	Asia Pacific University
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Micro, Small, and Medium Enterprises (MSMEs) represent 97.4% of all business establishments in Malaysia, contributing 38.4% of national GDP and employing over 7.3 million people. Despite their critical economic role, Malaysian SMEs face severe hurdles in digital transformation and automation. While the Malaysian government allocates hundreds of millions in matching grants across agencies like MDEC, SME Corp, MATRADE, and HRD Corp, a vast majority of these funds remain underutilized and unclaimed.

This problem stems from three systemic bottlenecks:

1. Low Awareness & Scattered Portals: Grant schemes are siloed across disconnected government portals with disparate application windows, making proactive discovery nearly impossible for lean SME teams.

2. Complex Multi-Variable Eligibility: Eligibility depends on strict, non-negotiable thresholds including Malaysian equity ratios (≥51% or ≥60%), sector-based headcount and turnover caps, operating tenure, and accredited Technology Service Provider (TSP) whitelists. Navigating these rules without expensive consultants leads to high disqualification rates.

3. Paperwork Fatigue: Drafting formal project justifications, compiling SSM documentation, and reconciling itemized invoices require days of manual effort that small business owners cannot spare.

Economically, this gap widens the digital divide, slowing national productivity and hindering Malaysia's progress under SDG 9 (Industry, Innovation, and Infrastructure). Small businesses continue absorbing 100% of modernization costs rather than leveraging matching funds specifically earmarked to support them.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SME GrantFlow is an autonomous, event-driven grant copilot and claim engine that inverts the traditional grant discovery model: grants find the SME, rather than the SME hunting through government portals. By integrating with everyday business workflows such as invoice ingestion, SSM profile updates, or scheduled opportunity scans. GrantFlow autonomously detects co-funding opportunities and prepares ready-to-submit application dossiers.

GrantFlow applies Artificial Intelligence through a hybrid, zero-hallucination architecture:

1. Unstructured Data Extraction & Narrative Drafting: Leveraging Generative AI APIs (DeepSeek cloud API and high-performance open-source models like DeepSeek-R1-Distill-Qwen-7B deployed on RunPod Serverless GPU), the AI parses unstructured invoice PDFs and crafts agency-compliant project justification narratives.

2. Deterministic Validation Boundary: Quantitative eligibility criteria (Malaysian equity percentages, operating tenure, SME Corp size caps, and MDEC approved TSP matching) are strictly evaluated by deterministic Python rule engines to prevent LLM hallucinations.

3. Human-in-the-Loop Supremacy: Business owners review transparent pass/fail breakdowns, edit AI-generated narrative drafts, and provide explicit sign-off before submission.

To deliver an enterprise-grade solution, our team leveraged key technical competencies gained from SDC Phase 1 workshops:

- Cloud Hosting & Prototyping: The FastAPI core backend is deployed on Replit, utilizing Replit's environment management, autoscale hosting, and secrets configuration, paired with a Next.js 16 frontend on Vercel.
- Modern Database Architecture: We implemented Neon Serverless PostgreSQL alongside local SQLite, featuring a tamper-evident, SHA-256 hash-chained immutable audit log that cryptographically validates every system event, actor, and submission packet.
- Scalable AI Compute: Serverless GPU orchestration via RunPod ensures low-latency, resilient fallback inference across cloud and local tiers. GrantFlow democratizes access to public capital, turning administrative friction into instant economic empowerment.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

1. Technologies, Frameworks, APIs & Databases

- Frontend: Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS 4, Lucide Icons, deployed on Vercel.
- Backend: Python 3.11, FastAPI, Pydantic v2 (strict schema validation), Uvicorn ASGI server, pytest test suite.
- Rules Engine: Custom deterministic Python rule modules (zero-hallucination compliance checking for MDEC, SME Corp, MATRADE, HRD Corp, MIDA, MITI, and Cradle).
- Database & Integrity: SQLite (authoritative local embedded database) paired with Neon Serverless PostgreSQL cloud mirror; cryptographic SHA-256 hash-chained immutable audit log.
- AI & LLM Orchestration: Multi-tier fallback pipeline:

- Tier 1: DeepSeek Cloud API (OpenAI-compatible format).

- Tier 2: RunPod Serverless GPU (vLLM endpoint running deepseek-ai/DeepSeek-R1-Distill-Qwen-7B).

- Tier 3: Deterministic rule-based template engine ensuring 100% offline availability.

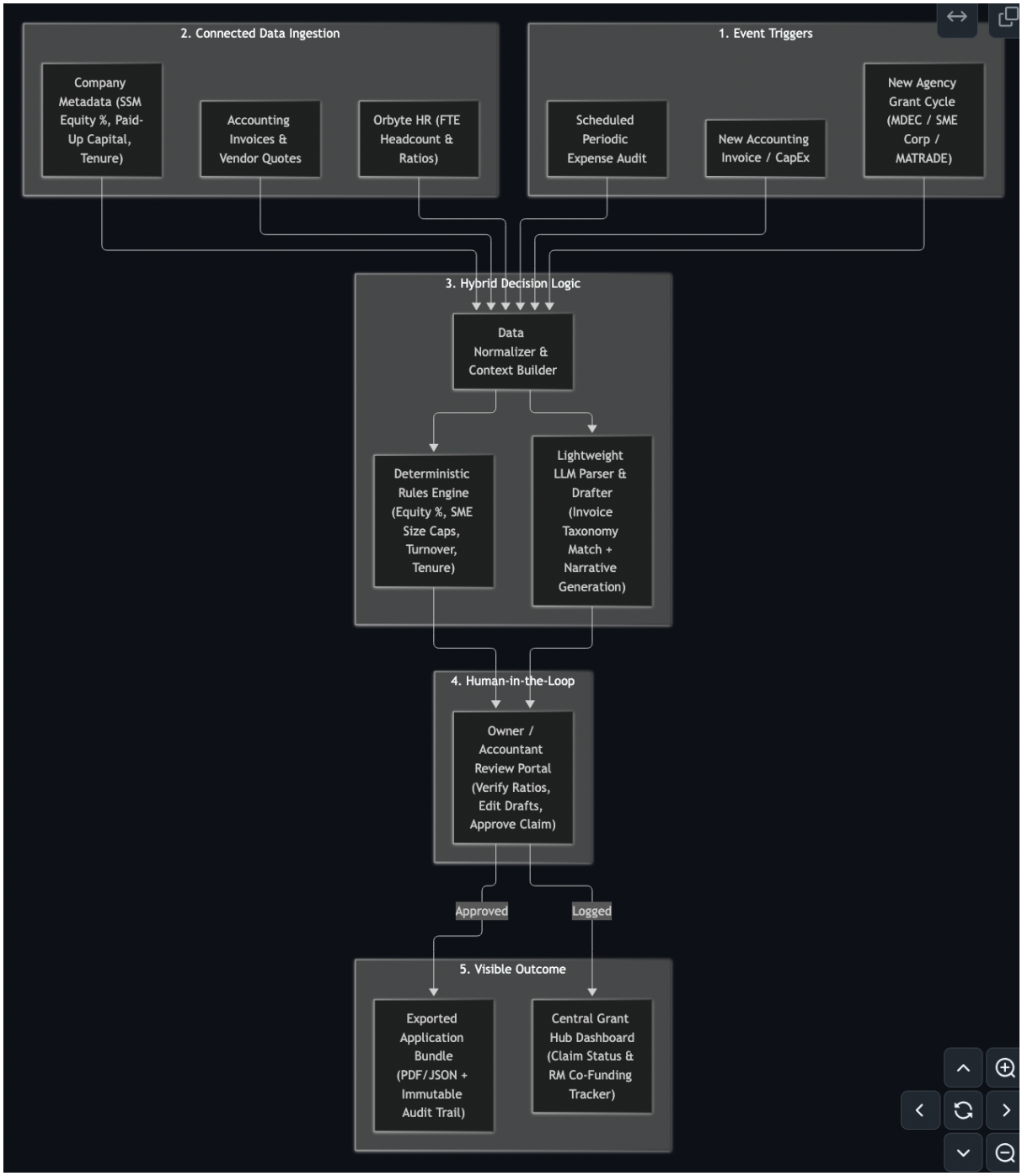
2. Utilization of SDC Sponsor / Cloud Platforms

Yes, our project actively leverages three official cloud and AI platforms:

- Replit: Used as the primary cloud hosting and development environment. The FastAPI backend is configured via replit for autoscale container deployment, providing secure environment/secrets management and continuous public API access for frontend integration.
- RunPod: Used as serverless GPU infrastructure. We deployed an OpenAI-compatible vLLM endpoint hosting deepseek-ai/DeepSeek-R1-Distill-Qwen-7B on dedicated GPU

compute to perform OCR parsing of unstructured invoices and draft formal grant justification narratives.

- Neon PostgreSQL: Utilized as the serverless cloud database for synchronizing SME profiles, claim records, and immutable audit logs with instant branching and scaling.



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied design thinking across five iterative stages:

1. Empathize & Define: Interviews with SME owners and accountants revealed that grant underutilization stems from fear of compliance disqualification, scattered portals, and tedious paperwork. Small businesses cannot afford dedicated grant consultants to track deadlines or decipher complex guidelines.

2. Ideate & Prototype: We devised a "zero-hallucination" architecture decoupling quantitative compliance from AI creativity. Deterministic Python rules evaluate strict criteria (Malaysian equity %, tenure, SME Corp turnover caps, approved TSP vendors), while GenAI parses unstructured invoices and drafts tailored project narratives.

3. User-Friendly Testing: We enforced Human-in-the-Loop Supremacy where owners review color-coded pass/fail checks, edit AI-generated narratives, and approve matching funds in one click, generating a tamper-proof SHA-256 sealed application packet.

4. Phase 2 Deployment Viability: Built on decoupled OpenAPI REST standards, GrantFlow operates seamlessly offline (SQLite) and cloud-scale (Neon PostgreSQL, Replit, RunPod). For Phase 2, the platform is architected to ingest live SSM registry data and LHDN MyInvois e-invoice streams via webhooks, enabling automated, audit-proof grant claiming at a national scale.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary Target: UN SDG Target 9.3: "Increase the access of small-scale industrial and other enterprises, in particular in developing countries, to financial services, including affordable credit, and their integration into value chains and markets." Secondary Target: UN SDG Target 9.b: "Support domestic technology development, research and innovation in developing countries, including by ensuring a conducive policy environment for, inter alia, industrial diversification and value addition to commodities."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Baseline 1 (Low Participation): Currently, less than 18% of eligible Malaysian MSMEs successfully tap into government digitalization and automation grant programs, leaving an estimated RM 300+ million in earmarked matching funds underutilized annually due to fragmented agency portals and documentation friction. Baseline 2 (Time & Effort Burden): Preparing a single grant submission manually requires an average of 14 to 21 business days (approx. 20–30 hours of manual reconciliation between business owners and external accounting consultants).

Metric Category	Your Response
	Baseline 3 (Rejection Rate): Up to 35% of initial submissions are returned or rejected due to avoidable compliance oversights (e.g., non-compliant equity thresholds, unaccredited TSP vendors, or mismatched sector caps). (Sources: SME Corporation Malaysia, SME Annual Report 2023/2024; World Bank, Malaysia Economic Monitor: Assessing MSME Access to Finance & Digitalization, 2023; DOSM MSME Performance 2024).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. Grant Application Turnaround Time: Elapsed time (in minutes/hours) from invoice/quotation ingestion to a validated, sign-off-ready application dossier. 2. Pre-Submission Compliance Accuracy Rate (%): Percentage of quantitative criteria verified without mathematical or threshold discrepancy prior to agency submission. 3. Grant Discovery Multiplier: Average number of viable co-funding schemes matched per SME profile across multi-agency portfolios (MDEC, SME Corp, MATRADE, HRD Corp, etc.). 4. Administrative Cost Savings (RM): Estimated consulting and accounting preparation fees saved per grant submission.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	1. >90% reduction in application preparation time: Slashing manual turnaround from 14 business days down to under 15 minutes of owner review and sign-off. 2. 100% elimination of quantitative eligibility errors: Achieving zero mathematical or threshold disqualifications via deterministic Python rule engines before packet generation. 3. 300% increase in discovered grant opportunities: Enabling multi-agency grant stacking where eligible SMEs identify an average of 2 to 3 qualifying programs per transaction instead of just 1.

Metric Category	Your Response
	4. 80% reduction in preparation overhead costs: Saving an estimated RM 2,000–RM 5,000 in third-party consultancy documentation fees per SME applicant.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Khoo

[Signature]

25/10/2026